MILITARY SPECIFICATION SHEET

COMPUTER SET, STANDARD AIRBORNE
CP-1430/AYK-14(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

The complete requirements for acquiring the computer described herein shall consist of this specification and the latest issue of MIL-C-85518.

REQUIREMENTS:

System configuration:
SRA location - See Table I.

TABLE I. SRA location in the CP-1430 computer.

SRA	Chassis Location	MIL-C-85518 Reference
DIOM	A01	3.6.2.2.1.7
-	A02	--
-	A03	--
PPSM	A04	3.6.2.2.1.6
PPSM	A05	3.6.2.2.1.6
PPSM	A06	3.6.2.2.1.6
-	A07	--
-	A08	--
SIM-A	A09	3.6.2.2.1.2
-	A10	--
IOP	A11	3.6.2.1.1
PSM	A12	3.6.2.1.1
GPM	A13	3.6.2.1.1
EAU	A14	3.6.2.1.4
MCM	A15	3.6.2.3.3
CMM-32	A16	3.6.2.3.1.1
CMM-32	A17	3.6.2.3.1.1
CMM-32	A18	3.6.2.3.1.1
CMM-32	A19	3.6.2.3.1.1
PCM-2	A20	3.6.2.4.2
Type 4 Chassis	-	3.2.2.5.1.3

Total weight: 61 lbs. maximum, excluding external cables.

Dimensions: Length 22.562 inches, width 10.237 inches, height 8.516 inches excluding front panel connectors and handles. See Figure 49 of MIL-C-85518.

Reliability in mean-time-between-failure (MTBF): MTBF shall be 615 hours minimum when tested in accordance with 4.4.3.4 of MIL-C-85518.

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Mean-time-to-repair (MTTR):
Intermediate level SRA replacement: MTTR shall be 24 minutes maximum.

Connector assignments: See Figure 1. Maintenance connector and all unused connectors shall be capped.

Connectors: Cables and connectors in accordance with MIL-E-5400.
Input/Output: In accordance with MIL-C-81511.
Input Power: In accordance with MIL-C-26482.
Maintenance: In accordance with MIL-C-38999

Service conditions: As specified in MIL-E-5400 for Class 2X equipment using conditioned air except altitude shall be limited to 50,000 feet as shown in Figure 2.

Cooling methods: Cooling shall be provided. If air vents are used, they shall be protected from the entry of harmful foreign materials. Aircraft structure shall not be used for heat sink without specific approval from the procuring activity.

Cooling air flow criteria:
Temperature: Cooling air at the inlet of this cooled equipment shall range between 35 to 120°F (1.7 to 48.9°C).
Ground cooling: Inlet temperature shall not exceed 120°F (48.9°C).
Flight cooling: Inlet temperature shall not exceed 80°F (26.7°C).

Cooling flow: Air flow for adequate cooling shall not exceed 2.2 pounds per minute, at 80°F (26.7°C) or 1.2 pounds per minute at 35°F (1.7°C). This equipment shall be structurally capable of withstanding cooling air flow two times greater than the maximum flow rate at sea level for 160°F (71°C) ambient; this factor decreases linearly with altitude to 1.5 times at 50,000 feet 100°F (38°C) ambient.

Pressure drop: The cooling air total pressure drop for the equipment is defined as the difference between the total pressure (static plus dynamic) entering the unit and the ambient pressure surrounding the unit. The total pressure drop of this unit shall be 1.42 (±10 percent) inches of water at the design flow rate corresponding to cooling air inlet temperature of 80°F (26.7°C) and an exit temperature of 160°F (71°C) at an ambient pressure of 14.7 psia. Pressure loss adjusting devices may be incorporated only if specific approval is obtained. The use of any selective devices, e.g., tailored orifices for pressure loss adjusting shall not be permitted.

Heat dissipation: Electrical heat generated within the unit (Qe) shall be 619 watts maximum. Heat load due to ambient conditions (Qa) shall be 123 watts maximum.

Surrounding air pressure: Equipment shall operate at 1.68 to 14.7 psia with surrounding air pressure varying at rates as high as 5.0 psi per minute.

Input power requirements: 115 ± 5 Volts ac, 400 ± 2 Hz (3 phase WYE), 619 Watts/730 VA maximum.

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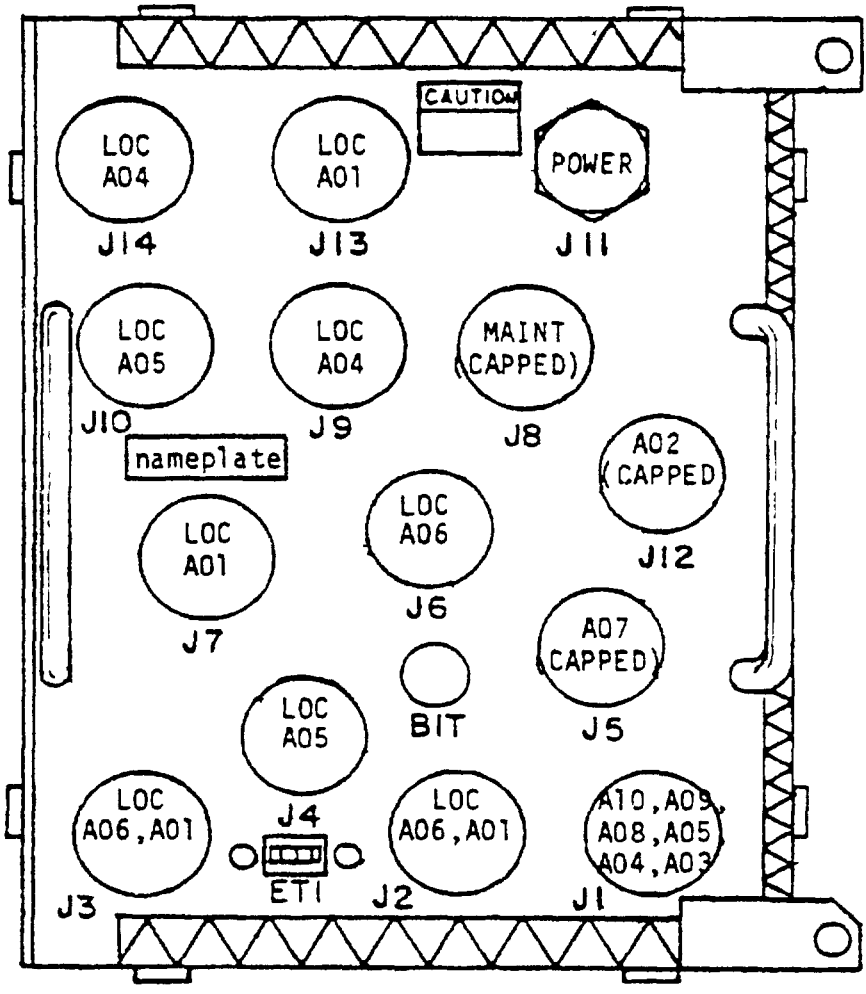


FIGURE 1. CP-1430 front panel.

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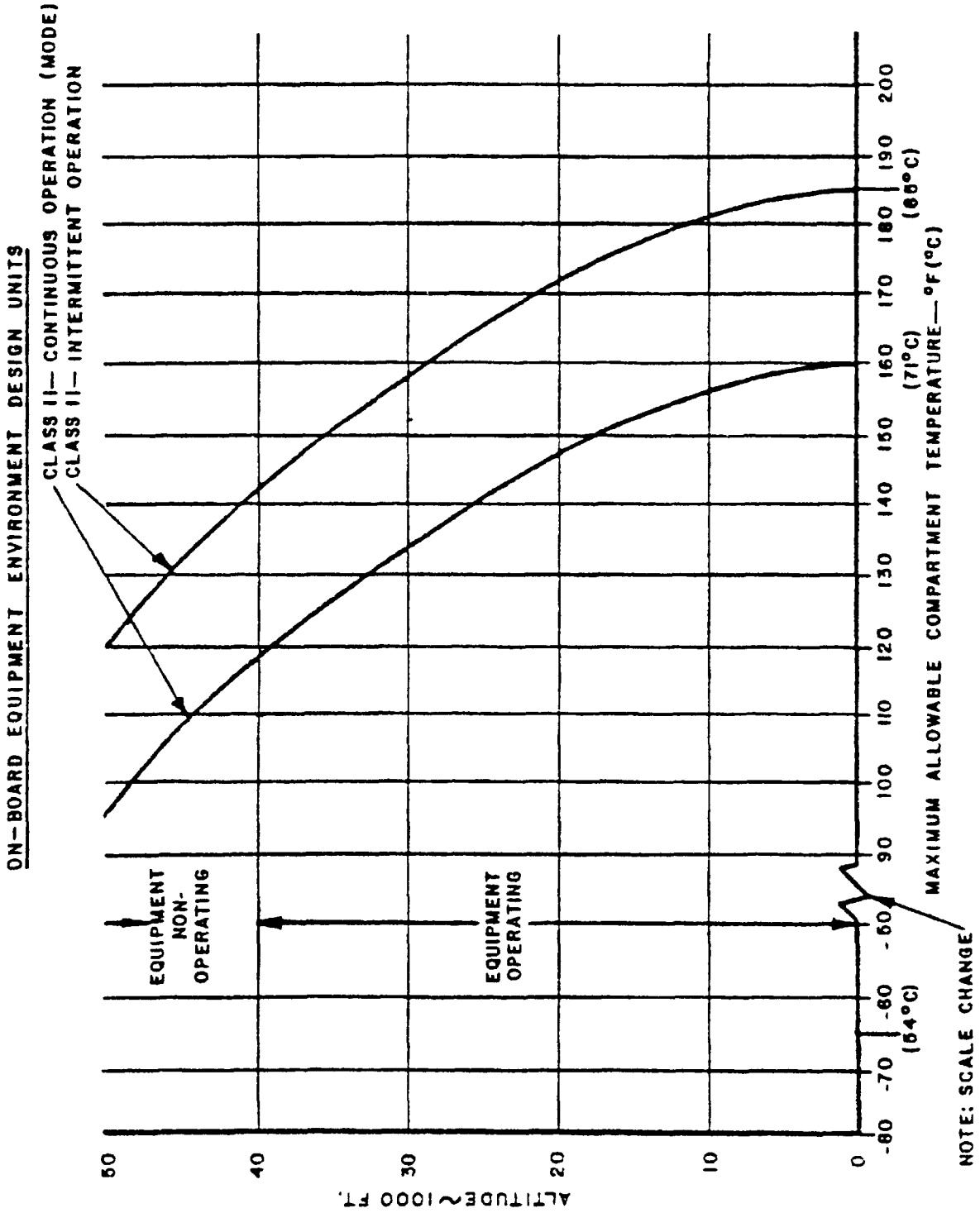


FIGURE 2. Temperature - altitude profile.

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Quality assurance provision: With the exception of AET, when cooling air is required during tests specified in 4.2, 4.3, and 4.4 of MIL-C-85518, the air shall be supplied at a temperature in the range of -65°F to 120°F. The flow rate shall be adjusted to correspond to the selected inlet temperatures, an exhaust temperature of 160°F, and the total heat load on the unit.

Preparing Activity
Navy AS

(Project 7021-N005-4

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